DEMO DIAGNOSTIC LABORATORY

Comprehensive Health Check-up • Synthetic report for RAG / Python practice

SAMPLE / NOT FOR CLINICAL USE

No real patient • No real diagnostic result

COMPREHENSIVE HEALTH CHECK-UP REPORT

Demo format inspired by common Indian diagnostic-lab reports; branding and patient data are fictional.

Patient Name	DEMO PATIENT (FICTIONAL)	Age / Gender	35 Years / Male
Patient ID	DEMO-RAG-2026-001	Sample Type	Blood + Urine
Collected	05/10/2026 08:30 AM	Reported	05/10/2026 05:30 PM
Report Status	Final / Cumulative	Purpose	RAG / NLP testing only

COMPLETE BLOOD COUNT (CBC)

Test	Result	Unit	Reference Range
Hemoglobin	14.2	g/dL	13.0–17.0
RBC Count	4.86	million/µL	4.5–5.9
Hematocrit (PCV)	42.8	%	40–50
MCV	88.1	fL	83–101
MCH	29.2	pg	27–32
MCHC	33.2	g/dL	31.5–34.5
RDW	13.4	%	11.5–14.5
WBC Count	7,420	/µL	4,000–11,000
Neutrophils	58	%	40–80
Lymphocytes	34	%	20–40
Monocytes	5	%	2–10
Eosinophils	3	%	1–6
Platelet Count	2.42	lakh/µL	1.5–4.5

DIABETES / GLUCOSE

Test	Result	Unit	Reference Range
Fasting Plasma Glucose	96	mg/dL	70–99
Post Meal Glucose	128	mg/dL	<140
HbA1c	5.6	%	<5.7
Estimated Average Glucose	114	mg/dL	—

KIDNEY FUNCTION / RENAL PROFILE

Test	Result	Unit	Reference Range
Urea	26	mg/dL	15–45
Creatinine, Serum	0.86	mg/dL	0.70–1.20
eGFR	108	mL/min/1.73m²	>90
Uric Acid	5.4	mg/dL	3.5–7.2
Sodium	139	mEq/L	135–145
Potassium	4.3	mEq/L	3.5–5.1
Chloride	101	mEq/L	98–107
Calcium	9.3	mg/dL	8.6–10.2
Phosphorus	3.7	mg/dL	2.5–4.5

LIVER FUNCTION TEST (LFT)

Test	Result	Unit	Reference Range
Total Bilirubin	0.72	mg/dL	0.2–1.2
Direct Bilirubin	0.18	mg/dL	0.0–0.3
Indirect Bilirubin	0.54	mg/dL	0.2–0.9

Test	Result	Unit	Reference Range
AST (SGOT)	27	U/L	0–40
ALT (SGPT)	29	U/L	0–41
GGT	24	U/L	9–48
Alkaline Phosphatase	82	U/L	40–129
Total Protein	7.3	g/dL	6.4–8.3
Albumin	4.5	g/dL	3.5–5.2
A/G Ratio	1.61	ratio	1.0–2.5

LIPID PROFILE

Test	Result	Unit	Reference Range
Total Cholesterol	176	mg/dL	<200
Triglycerides	112	mg/dL	<150
HDL Cholesterol	48	mg/dL	>40
LDL Cholesterol	106	mg/dL	<100 (optimal)
VLDL Cholesterol	22	mg/dL	5–40
Non-HDL Cholesterol	128	mg/dL	<130
Total Cholesterol / HDL	3.67	ratio	<5.0

THYROID PROFILE

Test	Result	Unit	Reference Range
TSH	2.18	µIU/mL	0.27–4.20
Free T3	3.2	pg/mL	2.0–4.4
Free T4	1.18	ng/dL	0.93–1.70

IRON STUDIES

Test	Result	Unit	Reference Range
Serum Iron	96	µg/dL	65–175
TIBC	318	µg/dL	250–450
Transferrin Saturation	30	%	20–50
Ferritin	86	ng/mL	30–400

VITAMINS

Test	Result	Unit	Reference Range
Vitamin B12	412	pg/mL	200–900
25-OH Vitamin D	31.6	ng/mL	30–100
Folate	8.7	ng/mL	4.6–18.7

URINE ROUTINE & MICROSCOPY

Test	Result	Unit	Reference Range
Colour	Pale Yellow	—	Pale Yellow
Appearance	Clear	—	Clear
Specific Gravity	1.018	—	1.005–1.030
pH	6.0	—	5.0–8.0
Protein	Negative	—	Negative
Glucose	Negative	—	Negative
Ketones	Negative	—	Negative
RBC	0–1	/HPF	0–2
WBC	1–2	/HPF	0–5
Epithelial Cells	Few	/HPF	Few

SYNTHETIC REPORT SUMMARY / RAG METADATA

Field	Value
Overall status	Sample values are mostly within the example reference ranges shown above.
Flagged example	LDL cholesterol 106 mg/dL is slightly above the displayed optimal target of <100 mg/dL.
Diabetes example	Fasting glucose 96 mg/dL and HbA1c 5.6% are within the displayed non-diabetic ranges.
Thyroid example	TSH, Free T3 and Free T4 are within the displayed reference ranges.
Kidney example	Creatinine 0.86 mg/dL and eGFR 108 mL/min/1.73m² are within the displayed example ranges.
Important	This is synthetic data created for software/RAG testing. It must not be used for medical diagnosis or treatment.

Suggested RAG questions for this document:

- What is the patient's HbA1c and what reference range is shown?
- What are the creatinine and eGFR values?
- List all lipid profile results with units.
- Which result is above the displayed optimal target?
- What are the TSH, Free T3 and Free T4 values?
- What are the Vitamin B12 and Vitamin D results?
- Which urine parameters are negative?
- Give a section-wise summary of the report.

Disclaimer: This PDF is a fictional educational document. It is not issued by, affiliated with, or endorsed by Dr Lal PathLabs or any diagnostic laboratory. All names, IDs, dates and test results are synthetic.